

Continuous Random Variables

Anaya Scott

✖ Question 1

✔ 0/1 pt ↻ 1 ↺ 19

Find $t_{0.85}$ for T distribution with 3 degrees of freedom.

$t_{0.85} =$ (Round the answer to 3 decimal places)

Question Help: [▶ Video](#)

● Question 2

✔ 0/1 pt ↻ 3 ↺ 19

Find $\chi^2_{0.005}$ for χ^2 distribution with 80 degrees of freedom.

$\chi^2_{0.005} =$ _____ (Round the answer to 3 decimal places)

Question Help: [▶ Video 1](#) [▶ Video 2](#)

● Question 3

✔ 0/1 pt ↻ 3 ↺ 19

A city metro train on the white line runs every 30 minutes during the time between 12 am and 6 am. Let X be the waiting time of a randomly selected passenger between 12 am and 6 am.

1. Describe the distribution of X :

$X \sim$ (a= , b=)

2. Use the random variable notation to express symbolically each of the following:

- An event in which the waiting time of a randomly selected passenger is less than 7.

- The probability of an event in which the waiting time of a randomly selected passenger is less than 12 is equal to 0.4.

- The probability of an event in which the waiting time of a randomly selected passenger is less than 22.

3. Use the above probability density curve to find the probability that it takes:

i. less than 7 minutes to wait for the train?

(round the answer to 2 decimal places)

ii. more than 22 minutes to wait for the train?

(round the answer to 2 decimal places)

iii. between 8 and 19 minutes to wait for the train?

(round the answer to 2 decimal places)

iv. more than 38 minutes to wait for the train?

(round the answer to 2 decimal places)

v. less than 40 minutes to wait for the train?

(round the answer to 2 decimal places)

4. Compute the following:

i. the expected value of X .

$$\mu_X = \underline{\hspace{2cm}} \text{ (round the answer to 2 decimal places)}$$

ii. the standard deviation of X .

$$\sigma_X = \underline{\hspace{2cm}} \text{ (round the answer to 2 decimal places)}$$

Question Help:  [Video](#)

● Question 4

 0/1 pt  3  19

A population of values has a normal distribution with $\mu = 241.1$ and $\sigma = 84$. You intend to draw a random sample of size $n = 123$.

What is the mean of the distribution of sample means?

$$\mu_{\bar{x}} = \underline{\hspace{2cm}}$$

What is the standard deviation of the distribution of sample means?
(Report answer accurate to 2 decimal places.)

$$\sigma_{\bar{x}} = \underline{\hspace{2cm}}$$



Question Help:  [Video](#)

● Question 5

 0/1 pt  3  19

When should you use t scores?

I. The population is normally distributed and the population standard deviation, σ , is known, regardless of the sample size.

II. The sample size is less than 30, and the population is not normally distributed.

III. The sample size is less than 30, the population is normally distributed and the population standard deviation is NOT known.

- ☐ II
- ☐ I and II
- ☐ I and III
- ☐ III
- ☐ I, II, and III
- ☐ II and III
- ☐ I

● Question 6

0/1 pt 3 19

According to a USPS study, the weight of a randomly selected shipment box is normally distributed with a mean of 19.2 lbs and a standard deviation of 4 lbs. Let X be the weight of a randomly selected shipment box and let S be the total weight of a random sample of size 14.

1. Describe the probability distribution of X and state its parameters μ and σ :

$X \sim$ ($\mu =$ _____, $\sigma =$ _____)

and find the probability that the weight of a randomly selected shipment box is less than 9 lbs.

_____ (Round the answer to 4 decimal places)

2. Use the Central Limit Theorem

to describe the probability distribution of S and state its parameters μ_S and σ_S :
(Round the answers to 1 decimal place)

$S \sim$ ($\mu_S =$ _____, $\sigma_S =$ _____)

and find the probability that the total weight of a sample of 14 randomly selected shipment boxes is less than 233 lbs.

_____ (Round the answer to 4 decimal places)

Question Help:  [Video](#)

✓ Question 7

✓ 1/1 pt ↻ 2 ↺ 19

The amount of detergent dispensed into bottles of liquid laundry detergent bottles for a particular brand is normally distributed with a mean of 80.9 ounces with a standard deviation of 0.8 ounces. If twenty-three bottles are randomly chosen from the factory, what is the probability that the mean fill is more than 81.5 ounces?

Round your answer to at least three decimal places.



If twenty-three randomly selected bottles had a mean of 81.5 oz, should the machine dispensing the detergent be recalibrated?



Hint 1

Question Help: [Video](#)

✓ **Question 8**

1/1 pt 2 19

Historically, heights of women have a mean of 159 cm and a standard deviation of 6.9 cm.

Use the three standard deviations rule to approximate the interval in which the majority of heights of women are in.



Sheniqua's height is 142.3 cm. Is her height an outlier?



Show Detailed Solution

Question Help: [Video](#)